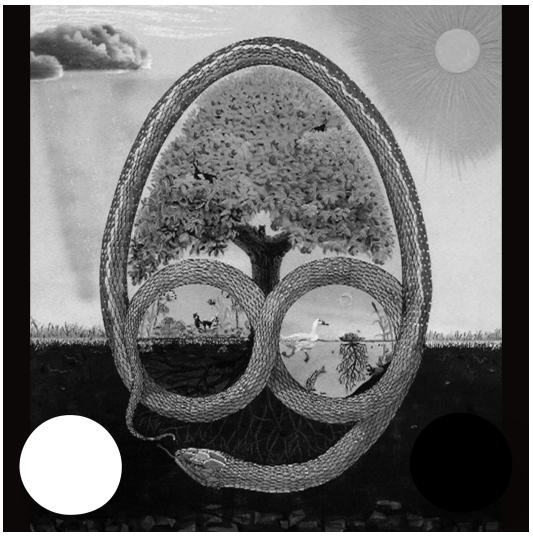


Merkle Providence Reverse RAG

Provenance-Preserving Cache Chains for Small Language Models

How Merkle trees turn Reverse RAG into a verifiable, shared knowledge layer.

How Hermes 3 Llama 3.1 8B answers with the confidence of a verified record, not a guess.

Russell Ballestrini <russell@unturf.com>, David Wong <david.s.wong@pm.me>, Riley Morgan <guybriley02@gmail.com>, Duck <duck.the.musician@gmail.com>

unfirehose.com · uncloseai.com · permacomputer.com

April 2026

Abstract

License: AGPL-3.0-only · This algorithm, its implementation, & all associated code carry the GNU Affero General Public License v3.0 (only). You may use, modify, & distribute under those terms. No proprietary relicensing exists.

Reverse Retrieval Augmented Generation (Reverse RAG) demonstrated that client-side context injection lets small 8B-parameter language models outperform much larger models on page-specific questions. Our reference implementation runs **Hermes 3**, a NousResearch instruction fine-tune of Meta's Llama 3.1 8B, served freely at uncloseai.com. Our prior work showed that the client already holds the document. No vector database required. No embedding pipeline. No retrieval failures.

Merkle Providence Reverse RAG extends this with a second insight: the client not only holds the document, it can *remember what it already computed about that document*, prove that memory came from an unmodified source, & share that verified knowledge with others on public or private chains.

A Merkle tree over document content chunks produces a root hash: a 32-byte fingerprint that changes when any part of the document changes. Pairing this root hash with a question hash yields a cache key. A cache hit means a previously computed answer returns instantly, with a Merkle proof of origin. A cache miss triggers fresh inference, stores the result, & extends our chain.

Unfirehose serves as our chain substrate. Every inference session produces JSONL records that feed into SQLite at `~/.unfirehose/unfirehose.db`. Public chains broadcast verified answer records to peers. Private chains stay local. Either way, a small model gains access to a growing body of pre-verified answers, each carrying cryptographic proof of the document version that produced it.

This combination removes two limits that have constrained small models:

1. **The repetition tax**: answering the same question about the same document on every visit, burning inference tokens & time, producing answers that may vary slightly each run.
2. **The trust deficit**: small models hallucinate. Cached answers with Merkle proofs do not hallucinate. They either match a verified record or they do not exist yet.

Once published, this technique forces a reckoning: every RAG system that cannot prove provenance of its retrieved chunks operates on unverifiable context. Merkle Providence makes provenance a first-class primitive.

Reference Model: Hermes 3 Llama 3.1 8B

Our reference implementation runs on **Hermes 3**, a fine-tune of Meta's Llama 3.1 8B developed by NousResearch. Hermes 3 builds on Llama 3.1's 128k context window & adds instruction alignment, function-calling discipline, & reasoning consistency that raw base models lack. NousResearch trained Hermes 3 to follow multi-turn conversation formats precisely, making it reliable for our structured providence cache interactions.

We chose Hermes 3 Llama 3.1 8B for three reasons:

Size matches our thesis. 8 billion parameters represent a model any person can run on consumer hardware. Our reference deployment runs Hermes 3 on a single RTX 3090 (24GB VRAM) at 8-bit quantization via bitsandbytes, consuming ~10GB VRAM with negligible quality loss over full precision. A MacBook M2 Pro runs it in CPU-offload mode. Our technique produces no value if it requires expensive hardware to demonstrate.

Fine-tuning demonstrates our point. Hermes 3 shows that fine-tuning a base model for instruction following, not scaling parameters, produces the reliability gains that matter for user-facing applications. Our Merkle Providence layer extends this logic: fine-tuned behavior + verified cached context beats raw parameter count.

uncloseai.com serves it freely. Our public inference endpoint at uncloseai.com runs Hermes 3 Llama 3.1 8B at no cost to any user. Our Reverse RAG pipeline in `uncloseai.js` targets this endpoint by default. Our Merkle Providence cache layer builds directly on this existing infrastructure. Any person anywhere, on any device with a browser, accesses our same Hermes 3 instance our paper describes.

Hermes 3 Llama 3.1 8B outperforms many larger models on page-specific questions when Reverse RAG supplies full context. With Merkle Providence caching, it delivers consistent, verifiable answers across sessions. Our paper uses Hermes 3 throughout all examples & benchmarks.

1. The Problem Reverse RAG Left Open

Reverse RAG solved context retrieval. Client-side DOM extraction replaced vector databases. Full-page injection replaced chunk fragmentation. Small models with perfect context outperformed large models with partial context.

One problem remained: **repetition**.

A user visits a technical document. Hermes 3 Llama 3.1 8B reads 8,000 tokens of page content, classifies the page, & answers their question. One hour later, a different user visits our same document. Our same Hermes 3 instance reads our same 8,000 tokens, classifies our same page, & answers a similar question again. Tomorrow, a third user. Our same 8,000 tokens. Our same classification. Our same inference cycle.

Nothing remembered. Nothing reused. No proof that any two answers came from our same source.

This repetition carries three costs:

- **Compute cost**: every visit re-runs inference on content that has not changed.
- **Consistency cost**: stochastic inference on identical inputs produces slightly different answers. Two users asking near-identical questions may receive meaningfully different responses from our same page.

- **Trust cost**: no mechanism exists to verify that an answer actually derived from a specific document version. A model can claim anything.

Merkle Providence addresses all three.

2. Merkle Trees as Content-Addressed Cache Keys

A Merkle tree splits a document into chunks, hashes each chunk, then builds a binary tree of hashes upward until a single root hash remains. This root hash has two properties that make it ideal for caching:

Property 1: Determinism. Our same document always produces our same root hash. Two clients processing our same page independently compute identical roots. No coordination required.

Property 2: Tamper sensitivity. Any change to any byte of the document changes at least one leaf hash, which propagates upward to invalidate our root. A document that changed since our last visit produces a new root hash automatically. Our cache never serves a stale answer for a modified document.

Cache key construction:

```
document_root = merkle_root(chunks(page_content))
question_key  = sha256(normalize(user_question))
cache_key     = document_root + ":" + question_key
```

A cache hit at this key means: *an earlier session computed an answer to this question, about this exact version of this document.* Our answer arrives in O(1) time. Our Merkle proof confirms our document version.

A cache miss means: run inference, store our result, extend our chain.

3. The Providence Layer: Proof of Origin

"Providence" carries two meanings that both apply here:

Provenance: where something came from. A Merkle proof lets any verifier confirm that an answer derived from a specific document version without receiving our full document. Our proof path, a sequence of sibling hashes from leaf to root, constitutes a compact cryptographic certificate of origin.

Foresight: the system appears to "already know" answers to questions asked before. From a user's perspective, a cache hit looks like instant comprehension. Our model responded before seemingly finishing to read. This resembles foresight because our prior sessions did our work ahead of time.

A providence record for a cached answer contains:

```
{
  "cache_key":      "sha256:abc123...:sha256:def456...",
  "document_root":  "sha256:abc123...",
  "document_uri":   "https://example.com/docs/api",
  "question_hash":  "sha256:def456...",
  "question_text":  "What does the rate_limit field mean?",
  "answer_text":    "The rate_limit field specifies...",
  "merkle_proof":   ["sha256:111...", "sha256:222...", ...],
  "model":          "hermes-3-llama-3.1-8b",
  "timestamp":      "2026-04-06T18:00:00Z",
  "chain":          "public"
}
```

Any recipient of this record can verify our Merkle proof against our document root. If our document root matches what they compute from our same URI, our answer carries full provenance. If their computation produces a different root, our document changed, our answer no longer applies to their version.

4. Unfirehose as Chain Substrate

Unfirehose collects JSONL session data from machine learning harnesses across a mesh of compute nodes. Every tool call, every model response, every session event flows into `~/.unfirehose/unfirehose.db` via our background ingestion worker.

Our Merkle providence layer extends this existing infrastructure with a providence cache table:

```
CREATE TABLE providence_cache (
  cache_key        TEXT PRIMARY KEY,
  document_root    TEXT NOT NULL,
  document_uri     TEXT NOT NULL,
  question_hash    TEXT NOT NULL,
  question_text    TEXT NOT NULL,
  answer_text      TEXT NOT NULL,
  merkle_proof     TEXT NOT NULL,  -- JSON array of sibling hashes
  model            TEXT NOT NULL,
  created_at       INTEGER NOT NULL,
  chain            TEXT NOT NULL DEFAULT 'private',
  peer_count       INTEGER NOT NULL DEFAULT 0
);

CREATE INDEX idx_providence_root ON providence_cache(document_root);
CREATE INDEX idx_providence_uri  ON providence_cache(document_uri);
```

Two chain modes:

Private chain: records stay local at `~/.unfirehose/`. Our personal session history. Our model learns from our own prior work on documents we visit repeatedly. Our chain belongs to us alone.

Public chain: records broadcast to peers via our unfirehose router daemon. Any node on our mesh that has processed our same document root can contribute answers. Any node that queries our same document root can receive pre-verified answers from others. Our chain grows with every participant.

Public chains do not share document content. They share proofs. Our Merkle proof proves an answer came from a specific document version without transmitting our document. Document content stays on our originating client, exactly as Reverse RAG intended.

5. How Small Models Punch Outside Their Weight Class

Parameter count governs a model's ability to reason from first principles. Context quality governs a model's ability to answer questions about specific documents. Verified memory governs a model's ability to give consistent, trustworthy answers over time.

Large models win on reasoning from first principles. Small models, given the right tools, win on everything else.

Merkle Providence Reverse RAG gives small models three tools:

Tool 1: Full context (from Reverse RAG). Our same page injection described in our prior paper. Our model receives every word, every heading, every entity, every code block of our document. Nothing retrieved by cosine similarity. Nothing fragmented. Our full document, live from our DOM.

Tool 2: Verified recall (from our providence cache). Before our model reads our document, we check our cache. A cache hit returns our answer in milliseconds with a Merkle proof. Our model delivers a verified answer faster than any large model can load our full context. Our small model "already knew" because our prior session did our work.

Tool 3: Peer-verified answers (from public chains). On a public chain, our small model benefits from every session any participant ran against our same document version. A document processed by 100 sessions across our mesh produces a rich cache. Our Hermes 3 Llama 3.1 8B instance answers from our collective record, not just its own prior work.

These three tools operate at different timescales:

- **O(1)**: cache hit, local or peer. Instant. No inference required.

- **O(context)**: Reverse RAG with full page injection. Fast. One inference pass over complete context.
- **O(reasoning)**: fresh inference on novel questions. Standard. One inference pass, uncached.

A large model operates exclusively at O(reasoning). Our small model with Merkle Providence operates at O(1) for previously answered questions, O(context) for new questions on known documents, & O(reasoning) only for truly novel questions on new documents.

6. Architecture

Stage 1: Document Fingerprinting

Our same DOM extraction from Reverse RAG runs first. Once our extraction completes, we split our content into 512-token chunks & build our Merkle tree:

```
function merkleRoot(chunks) {
    let layer = chunks.map(c => sha256(c));
    while (layer.length > 1) {
        const next = [];
        for (let i = 0; i < layer.length; i += 2) {
            const left  = layer[i];
            const right = layer[i + 1] ?? left;  // odd node doubles itself
            next.push(sha256(left + right));
        }
        layer = next;
    }
    return layer[0];
}
```

Our root hash serves as our document fingerprint for our entire session. One hash computation. No server calls.

Stage 2: Cache Lookup

Before any LLM call, we check our providence cache:

```
async function lookupProvidence(documentRoot, questionText) {
    const questionHash = sha256(normalize(questionText));
    const cacheKey     = `${documentRoot}:${questionHash}`;

    // Check local chain first
    const local = await db.get(
        'SELECT * FROM providence_cache WHERE cache_key = ?',
        [cacheKey]
    );
    if (local) return { answer: local.answer_text, proof: local.merkle_proof, source: 'local' };

    // Check public chain peers
    if (publicChainEnabled) {
        const peer = await queryPeers(cacheKey);
        if (peer && verifyMerkleProof(peer.merkle_proof, documentRoot)) {
            await storeLocal(peer);  // cache locally for future hits
            return { answer: peer.answer_text, proof: peer.merkle_proof, source: 'peer' };
        }
    }

    return null;  // cache miss, proceed to inference
}
```

Stage 3: Inference & Cache Write

On a cache miss, our standard Reverse RAG inference runs. Once our answer arrives, we store our providence record:

```
async function storeProvidence(documentRoot, documentUri, question, answer, model) {
    const questionHash = sha256(normalize(question));
    const cacheKey     = `${documentRoot}:${questionHash}`;
    const merkleProof  = computeProof(documentRoot, question);  // path to root

    await db.run(`
        INSERT OR REPLACE INTO providence_cache
        (cache_key, document_root, document_uri, question_hash, question_text,
         answer_text, merkle_proof, model, created_at, chain)
        VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
    `, [cacheKey, documentRoot, documentUri, questionHash, question,
        answer, JSON.stringify(merkleProof), model, Date.now(), chainMode]);

    if (chainMode === 'public') broadcastToMesh(cacheKey);
}
```

Stage 4: Proof Verification

Any party that receives a providence record can verify it:

```
function verifyMerkleProof(proof, claimedRoot, leafContent) {
    let current = sha256(leafContent);
    for (const sibling of proof) {
        current = sha256(current < sibling
            ? current + sibling
            : sibling + current);
    }
    return current === claimedRoot;
}
```

If our document root computed from our verifier's live DOM matches `claimedRoot`, our answer carries full provenance. Our answer derives from our exact document version our verifier currently views. If our roots diverge, our document changed. Our cache entry no longer applies.

7. Public vs. Private Chains

Property	Private Chain	Public Chain
Storage	`~/.unfirehose/` local only	Broadcast via unfirehose router mesh
Participants	One user, one machine	Any node on our mesh
Document content shared	Never	Never (proofs only)
Answer content shared	Never	Yes, with Merkle proof
Trust model	Self-trust: our own prior work	Peer-trust: verified by Merkle proof
Best for	Personal document libraries, sensitive internal docs	Open documentation, public research, shared reference material
Cache growth rate	Linear in our own sessions	Linear in mesh participant sessions

Private chains suit corporate knowledge bases, personal research notes, any document our user does not want to share. Our local model benefits from our own prior sessions. No data leaves our machine.

Public chains suit open documentation, public APIs, research papers, any document where sharing verified answers benefits our community. No document content leaves our originating client. Proofs travel instead.

Our unfirehose router daemon manages chain membership & peer discovery. Our user opts in to public chains per-document-domain or per-session.

8. Forcing the Issue

Some technical formulations compel structural change not because they require adoption but because their existence makes our prior approach indefensible.

Our Merkle Providence layer forces one question onto every RAG system: *can you prove where your context came from?*

Traditional RAG retrieves chunks by cosine similarity. No chunk carries a Merkle proof. No retrieval carries a document fingerprint. Our retrieved context could come from a stale index, a corrupted chunk, or a document that changed since our last ingestion. Our RAG pipeline trusts our embedding similarity as a proxy for relevance. Nothing verifies our source.

Once Merkle-proved answers exist in our ecosystem, unverified RAG answers carry an implicit disclaimer: *I retrieved this from somewhere, I believe it matches your question, but I cannot prove our source was our document you currently view.*

Our providence cache answers carry a different posture: *I computed this answer from our exact bytes of our document whose root hash I can show you. Verify it yourself.*

This posture change does not require every system to adopt Merkle trees immediately. It requires every system to acknowledge that provenance matters, & that systems which cannot demonstrate provenance operate at a disadvantage in trust-sensitive contexts.

Medical documentation. Legal reference material. Security advisories. Financial disclosures. Any domain where "I retrieved something similar" falls short of "I can prove this came from our exact source you cited."

9. Unfirehose Integration

Unfirehose already collects JSONL from Claude Code, Fetch, & uncloseai harnesses across our mesh. Our providence cache extends this with three new API routes:

`GET /api/providence?uri=...`

Returns all cached answer records for a document URI. Includes document root hash, question hashes, answer previews, & peer counts. Lets users audit what our mesh knows about a document.

`POST /api/providence`

Stores a new providence record. Accepts `{ document_root, document_uri, question_text, answer_text, merkle_proof, model }`. Validates our proof before storing.

`GET /api/providence/peers?root=...`

Queries our mesh for cached records matching a document root hash. Returns answers from public-chain peers with their Merkle proofs. Our uncloseai.js client calls this before any inference to check for peer cache hits.

Our existing unfirehose dashboard gains a Providence page: a view of our cache landscape across documents, peer contribution rates, cache hit rates per domain, & our chain health.

10. Relationship to Zero-Knowledge Proofs

A Merkle proof constitutes a specific form of zero-knowledge argument: *I can prove this leaf exists in this tree without revealing our other leaves.*

In our context: *I can prove this answer came from this document version without revealing our full document.*

Our full ZKP apparatus (zk-SNARKs, Bulletproofs) provides stronger guarantees & more complex proofs. Merkle proofs provide a simpler, faster, & more practical subset: membership proofs sufficient for our provenance use case.

Our connection to ZKP deepens when our public chain handles sensitive documents. A user could prove their answer derived from a specific confidential document without revealing our document content, our question text, or our answer text. Only our proof path & our document root travel over our network. Our verifier confirms our proof without seeing our data.

This positions our public chain as opt-in verifiable knowledge sharing, not compelled disclosure.

11. Why This Matters for Permacomputer

Permacomputer infrastructure grows knowledge that outlasts our individual sessions, our individual machines, & our individual participants. Code seeds sprout on any abandoned technology. Knowledge should do our same.

A private chain that lives only on one machine dies with that machine. Our public chain on our unfirehose mesh survives node failures because our same document root on a different node still matches our cached proofs on our other nodes. Our knowledge persists as long as any mesh participant holds a copy.

Our Merkle structure aligns with permacomputer values:

Truth: proofs over assertions. Our answer carries a mathematical certificate, not just a model's confidence score.

Freedom: private chains for sensitive knowledge, public chains for shared knowledge. Our user chooses. No platform extracts our knowledge without consent.

Harmony: cache hits replace repeated inference. Our same compute produces more answers over time. Our mesh grows more efficient as our cache fills.

Love: small models with providence caches remove our infrastructure barrier. Hermes 3 Llama 3.1 8B on a consumer GPU, connected to our public chain, answers with our collective memory of our mesh. Our permacomputer does not require our most expensive hardware to deliver our most reliable answers.

12. Implementation Roadmap

Phase 1: Local private chain (unfirehose extension)

- Add `providence_cache` table to our unfirehose SQLite schema
- Extend our uncloseai.js Reverse RAG pipeline with pre-inference cache lookup & post-inference cache write
- Add `GET /api/providence` & `POST /api/providence` routes to our unfirehose web app
- Add Providence view to our dashboard

Phase 2: Public chain via unfirehose router

- Extend our router daemon with `/providence/broadcast` & `/providence/query` endpoints
- Add peer discovery to our mesh probe logic
- Add per-domain opt-in controls to our settings page
- Add peer contribution metrics to our Providence dashboard view

Phase 3: Proof verification in browser

- Implement `verifyMerkleProof()` in our uncloseai.js client
- Surface proof status in our chat UI (verified badge vs. fresh inference)
- Add document root display so users can audit our source fingerprint

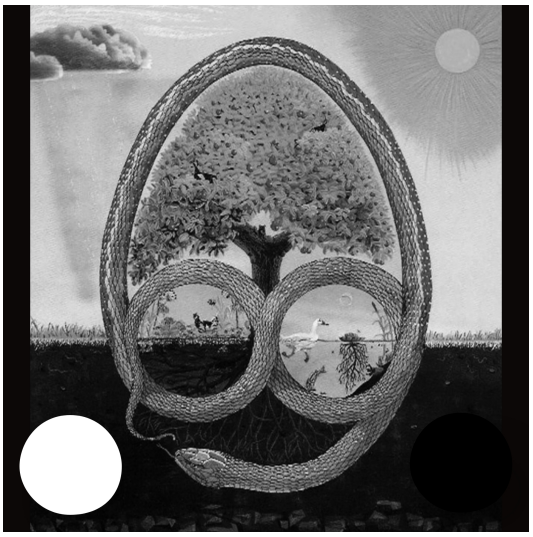
Acknowledgements

Our Merkle cache idea emerged from a conversation on X Spaces. We thank **@AR_Milne** & **@without_rulers** for a discussion that sparked the core insight: that a Merkle tree over document content could serve as a fast, verifiable cache layer for machine learning inference. Ideas born in public conversation belong to our commons.

Citation

```
Russell Ballestrini, David Wong, Riley Morgan, Duck.
"Merkle Providence Reverse RAG: Provenance-Preserving Cache Chains
for Small Language Models."
unfirehose.com, 2026.
https://unfirehose.com/merkle-providence-reverse-rag.html
```

License



GNU Affero General Public License v3

```
AGPL-3.0-only
```

PERMACOMPUTER PREAMBLE - NO WARRANTY

This is free software for the public good of a permacomputer hosted at permacomputer.com - an always-on computer by the people, for the people. One which is durable, easy to repair, and distributed like tap water for machine learning intelligence.

The permacomputer is community-owned infrastructure optimized around four values:

  TRUTH   - Source code must be open source & freely distributed
  FREEDOM - Voluntary participation without corporate control
  HARMONY - Systems operating with minimal waste that self-renew
  LOVE   - Individual rights protected while fostering cooperation

This paper contributes to that vision by documenting Merkle Providence Reverse RAG: a provenance-preserving cache layer that lets small language models punch far outside their parameter class. Code is seeds to sprout on any abandoned technology.

Learn more: https://www.permacomputer.com

NO WARRANTY. THE SOFTWARE IS PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND.

Copyright (C) 2026 Russell Ballestrini & David Wong & Riley Morgan & Duck
russell@unturf.com
david.s.wong@pm.me
guybriley02@gmail.com
duck.the.musician@gmail.com
https://www.unturf.com/software
https://www.permacomputer.com
https://uncloseai.com
https://unfirehose.com
https://russell.ballestrini.net

GNU AFFERO GENERAL PUBLIC LICENSE
Version 3, 19 November 2007

Copyright (C) 2007 Free Software Foundation, Inc. <https://fsf.org/>
Everyone is permitted to copy and distribute verbatim copies
of this license document, but changing it is not allowed.

Preamble

The GNU Affero General Public License is a free, copyleft license for software and other kinds of works, specifically designed to ensure cooperation with the community in the case of network server software.

The licenses for most software and other practical works are designed to take away your freedom to share and change the works. By contrast, our General Public Licenses are intended to guarantee your freedom to share and change all versions of a program--to make sure it remains free software for all its users.

When we speak of free software, we are referring to freedom, not price. Our General Public Licenses are designed to make sure that you have the freedom to distribute copies of free software (and charge for them if you wish), that you receive source code or can get it if you want it, that you can change the software or use pieces of it in new free programs, and that you know you can do these things.

Developers that use our General Public Licenses protect your rights with two steps: (1) assert copyright on the software, and (2) offer you this License which gives you legal permission to copy, distribute and/or modify the software.

A secondary benefit of defending all users' freedom is that improvements made in alternate versions of the program, if they receive widespread use, become available for other developers to incorporate. Many developers of free software are heartened and

encouraged by the resulting cooperation. However, in the case of software used on network servers, this result may fail to come about. The GNU General Public License permits making a modified version and letting the public access it on a server without ever releasing its source code to the public.

The GNU Affero General Public License is designed specifically to ensure that, in such cases, the modified source code becomes available to the community. It requires the operator of a network server to provide the source code of the modified version running there to the users of that server. Therefore, public use of a modified version, on a publicly accessible server, gives the public access to the source code of the modified version.

An older license, called the Affero General Public License and published by Affero, was designed to accomplish similar goals. This is a different license, not a version of the Affero GPL, but Affero has released a new version of the Affero GPL which permits relicensing under this license.

The precise terms and conditions for copying, distribution and modification follow.

TERMS AND CONDITIONS

0. Definitions.

"This License" refers to version 3 of the GNU Affero General Public License.

"Copyright" also means copyright-like laws that apply to other kinds of works, such as semiconductor masks.

"The Program" refers to any copyrightable work licensed under this License. Each licensee is addressed as "you". "Licensees" and "recipients" may be individuals or organizations.

To "modify" a work means to copy from or adapt all or part of the work in a fashion requiring copyright permission, other than the making of an exact copy. The resulting work is called a "modified version" of the earlier work or a work "based on" the earlier work.

A "covered work" means either the unmodified Program or a work based on the Program.

To "propagate" a work means to do anything with it that, without permission, would make you directly or secondarily liable for infringement under applicable copyright law, except executing it on a computer or modifying a private copy. Propagation includes copying, distribution (with or without modification), making available to the public, and in some countries other activities as well.

To "convey" a work means any kind of propagation that enables other parties to make or receive copies. Mere interaction with a user through a computer network, with no transfer of a copy, is not conveying.

An interactive user interface displays "Appropriate Legal Notices" to the extent that it includes a convenient and prominently visible feature that (1) displays an appropriate copyright notice, and (2) tells the user that there is no warranty for the work (except to the extent that warranties are provided), that licensees may convey the work under this License, and how to view a copy of this License. If the interface presents a list of user commands or options, such as a menu, a prominent item in the list meets this criterion.

1. Source Code.

The "source code" for a work means the preferred form of the work for making modifications to it. "Object code" means any non-source form of a work.

A "Standard Interface" means an interface that either is an official

standard defined by a recognized standards body, or, in the case of interfaces specified for a particular programming language, one that is widely used among developers working in that language.

The "System Libraries" of an executable work include anything, other than the work as a whole, that (a) is included in the normal form of packaging a Major Component, but which is not part of that Major Component, and (b) serves only to enable use of the work with that Major Component, or to implement a Standard Interface for which an implementation is available to the public in source code form. A "Major Component", in this context, means a major essential component (kernel, window system, and so on) of the specific operating system (if any) on which the executable work runs, or a compiler used to produce the work, or an object code interpreter used to run it.

The "Corresponding Source" for a work in object code form means all the source code needed to generate, install, and (for an executable work) run the object code and to modify the work, including scripts to control those activities. However, it does not include the work's System Libraries, or general-purpose tools or generally available free programs which are used unmodified in performing those activities but which are not part of the work. For example, Corresponding Source includes interface definition files associated with source files for the work, and the source code for shared libraries and dynamically linked subprograms that the work is specifically designed to require, such as by intimate data communication or control flow between those subprograms and other parts of the work.

The Corresponding Source need not include anything that users can regenerate automatically from other parts of the Corresponding Source.

The Corresponding Source for a work in source code form is that same work.

2. Basic Permissions.

All rights granted under this License are granted for the term of copyright on the Program, and are irrevocable provided the stated conditions are met. This License explicitly affirms your unlimited permission to run the unmodified Program. The output from running a covered work is covered by this License only if the output, given its content, constitutes a covered work. This License acknowledges your rights of fair use or other equivalent, as provided by copyright law.

You may make, run and propagate covered works that you do not convey, without conditions so long as your license otherwise remains in force. You may convey covered works to others for the sole purpose of having them make modifications exclusively for you, or provide you with facilities for running those works, provided that you comply with the terms of this License in conveying all material for which you do not control copyright. Those thus making or running the covered works for you must do so exclusively on your behalf, under your direction and control, on terms that prohibit them from making any copies of your copyrighted material outside their relationship with you.

Conveying under any other circumstances is permitted solely under the conditions stated below. Sublicensing is not allowed; section 10 makes it unnecessary.

3. Protecting Users' Legal Rights From Anti-Circumvention Law.

No covered work shall be deemed part of an effective technological measure under any applicable law fulfilling obligations under article 11 of the WIPO copyright treaty adopted on 20 December 1996, or similar laws prohibiting or restricting circumvention of such measures.

When you convey a covered work, you waive any legal power to forbid circumvention of technological measures to the extent such circumvention is effected by exercising rights under this License with respect to

the covered work, and you disclaim any intention to limit operation or modification of the work as a means of enforcing, against the work's users, your or third parties' legal rights to forbid circumvention of technological measures.

4. Conveying Verbatim Copies.

You may convey verbatim copies of the Program's source code as you receive it, in any medium, provided that you conspicuously and appropriately publish on each copy an appropriate copyright notice; keep intact all notices stating that this License and any non-permissive terms added in accord with section 7 apply to the code; keep intact all notices of the absence of any warranty; and give all recipients a copy of this License along with the Program.

You may charge any price or no price for each copy that you convey, and you may offer support or warranty protection for a fee.

5. Conveying Modified Source Versions.

You may convey a work based on the Program, or the modifications to produce it from the Program, in the form of source code under the terms of section 4, provided that you also meet all of these conditions:

a) The work must carry prominent notices stating that you modified it, and giving a relevant date.

b) The work must carry prominent notices stating that it is released under this License and any conditions added under section 7. This requirement modifies the requirement in section 4 to "keep intact all notices".

c) You must license the entire work, as a whole, under this License to anyone who comes into possession of a copy. This License will therefore apply, along with any applicable section 7 additional terms, to the whole of the work, and all its parts, regardless of how they are packaged. This License gives no permission to license the work in any other way, but it does not invalidate such permission if you have separately received it.

d) If the work has interactive user interfaces, each must display Appropriate Legal Notices; however, if the Program has interactive interfaces that do not display Appropriate Legal Notices, your work need not make them do so.

A compilation of a covered work with other separate and independent works, which are not by their nature extensions of the covered work, and which are not combined with it such as to form a larger program, in or on a volume of a storage or distribution medium, is called an "aggregate" if the compilation and its resulting copyright are not used to limit the access or legal rights of the compilation's users beyond what the individual works permit. Inclusion of a covered work in an aggregate does not cause this License to apply to the other parts of the aggregate.

6. Conveying Non-Source Forms.

You may convey a covered work in object code form under the terms of sections 4 and 5, provided that you also convey the machine-readable Corresponding Source under the terms of this License, in one of these ways:

a) Convey the object code in, or embodied in, a physical product (including a physical distribution medium), accompanied by the Corresponding Source fixed on a durable physical medium customarily used for software interchange.

b) Convey the object code in, or embodied in, a physical product (including a physical distribution medium), accompanied by a written offer, valid for at least three years and valid for as long as you offer spare parts or customer support for that product

model, to give anyone who possesses the object code either (1) a copy of the Corresponding Source for all the software in the product that is covered by this License, on a durable physical medium customarily used for software interchange, for a price no more than your reasonable cost of physically performing this conveying of source, or (2) access to copy the Corresponding Source from a network server at no charge.

c) Convey individual copies of the object code with a copy of the written offer to provide the Corresponding Source. This alternative is allowed only occasionally and noncommercially, and only if you received the object code with such an offer, in accord with subsection 6b.

d) Convey the object code by offering access from a designated place (gratis or for a charge), and offer equivalent access to the Corresponding Source in the same way through the same place at no further charge. You need not require recipients to copy the Corresponding Source along with the object code. If the place to copy the object code is a network server, the Corresponding Source may be on a different server (operated by you or a third party) that supports equivalent copying facilities, provided you maintain clear directions next to the object code saying where to find the Corresponding Source. Regardless of what server hosts the Corresponding Source, you remain obligated to ensure that it is available for as long as needed to satisfy these requirements.

e) Convey the object code using peer-to-peer transmission, provided you inform other peers where the object code and Corresponding Source of the work are being offered to the general public at no charge under subsection 6d.

A separable portion of the object code, whose source code is excluded from the Corresponding Source as a System Library, need not be included in conveying the object code work.

A "User Product" is either (1) a "consumer product", which means any tangible personal property which is normally used for personal, family, or household purposes, or (2) anything designed or sold for incorporation into a dwelling. In determining whether a product is a consumer product, doubtful cases shall be resolved in favor of coverage. For a particular product received by a particular user, "normally used" refers to a typical or common use of that class of product, regardless of the status of the particular user or of the way in which the particular user actually uses, or expects or is expected to use, the product. A product is a consumer product regardless of whether the product has substantial commercial, industrial or non-consumer uses, unless such uses represent the only significant mode of use of the product.

"Installation Information" for a User Product means any methods, procedures, authorization keys, or other information required to install and execute modified versions of a covered work in that User Product from a modified version of its Corresponding Source. The information must suffice to ensure that the continued functioning of the modified object code is in no case prevented or interfered with solely because modification has been made.

If you convey an object code work under this section in, or with, or specifically for use in, a User Product, and the conveying occurs as part of a transaction in which the right of possession and use of the User Product is transferred to the recipient in perpetuity or for a fixed term (regardless of how the transaction is characterized), the Corresponding Source conveyed under this section must be accompanied by the Installation Information. But this requirement does not apply if neither you nor any third party retains the ability to install modified object code on the User Product (for example, the work has been installed in ROM).

The requirement to provide Installation Information does not include a requirement to continue to provide support service, warranty, or updates for a work that has been modified or installed by the recipient, or for

the User Product in which it has been modified or installed. Access to a network may be denied when the modification itself materially and adversely affects the operation of the network or violates the rules and protocols for communication across the network.

Corresponding Source conveyed, and Installation Information provided, in accord with this section must be in a format that is publicly documented (and with an implementation available to the public in source code form), and must require no special password or key for unpacking, reading or copying.

7. Additional Terms.

"Additional permissions" are terms that supplement the terms of this License by making exceptions from one or more of its conditions. Additional permissions that are applicable to the entire Program shall be treated as though they were included in this License, to the extent that they are valid under applicable law. If additional permissions apply only to part of the Program, that part may be used separately under those permissions, but the entire Program remains governed by this License without regard to the additional permissions.

When you convey a copy of a covered work, you may at your option remove any additional permissions from that copy, or from any part of it. (Additional permissions may be written to require their own removal in certain cases when you modify the work.) You may place additional permissions on material, added by you to a covered work, for which you have or can give appropriate copyright permission.

Notwithstanding any other provision of this License, for material you add to a covered work, you may (if authorized by the copyright holders of that material) supplement the terms of this License with terms:

a) Disclaiming warranty or limiting liability differently from the terms of sections 15 and 16 of this License; or

b) Requiring preservation of specified reasonable legal notices or author attributions in that material or in the Appropriate Legal Notices displayed by works containing it; or

c) Prohibiting misrepresentation of the origin of that material, or requiring that modified versions of such material be marked in reasonable ways as different from the original version; or

d) Limiting the use for publicity purposes of names of licensors or authors of the material; or

e) Declining to grant rights under trademark law for use of some trade names, trademarks, or service marks; or

f) Requiring indemnification of licensors and authors of that material by anyone who conveys the material (or modified versions of it) with contractual assumptions of liability to the recipient, for any liability that these contractual assumptions directly impose on those licensors and authors.

All other non-permissive additional terms are considered "further restrictions" within the meaning of section 10. If the Program as you received it, or any part of it, contains a notice stating that it is governed by this License along with a term that is a further restriction, you may remove that term. If a license document contains a further restriction but permits relicensing or conveying under this License, you may add to a covered work material governed by the terms of that license document, provided that the further restriction does not survive such relicensing or conveying.

If you add terms to a covered work in accord with this section, you must place, in the relevant source files, a statement of the additional terms that apply to those files, or a notice indicating where to find the applicable terms.

Additional terms, permissive or non-permissive, may be stated in the form of a separately written license, or stated as exceptions; the above requirements apply either way.

8. Termination.

You may not propagate or modify a covered work except as expressly provided under this License. Any attempt otherwise to propagate or modify it is void, and will automatically terminate your rights under this License (including any patent licenses granted under the third paragraph of section 11).

However, if you cease all violation of this License, then your license from a particular copyright holder is reinstated (a) provisionally, unless and until the copyright holder explicitly and finally terminates your license, and (b) permanently, if the copyright holder fails to notify you of the violation by some reasonable means prior to 60 days after the cessation.

Moreover, your license from a particular copyright holder is reinstated permanently if the copyright holder notifies you of the violation by some reasonable means, this is the first time you have received notice of violation of this License (for any work) from that copyright holder, and you cure the violation prior to 30 days after your receipt of the notice.

Termination of your rights under this section does not terminate the licenses of parties who have received copies or rights from you under this License. If your rights have been terminated and not permanently reinstated, you do not qualify to receive new licenses for the same material under section 10.

9. Acceptance Not Required for Having Copies.

You are not required to accept this License in order to receive or run a copy of the Program. Ancillary propagation of a covered work occurring solely as a consequence of using peer-to-peer transmission to receive a copy likewise does not require acceptance. However, nothing other than this License grants you permission to propagate or modify any covered work. These actions infringe copyright if you do not accept this License. Therefore, by modifying or propagating a covered work, you indicate your acceptance of this License to do so.

10. Automatic Licensing of Downstream Recipients.

Each time you convey a covered work, the recipient automatically receives a license from the original licensors, to run, modify and propagate that work, subject to this License. You are not responsible for enforcing compliance by third parties with this License.

An "entity transaction" is a transaction transferring control of an organization, or substantially all assets of one, or subdividing an organization, or merging organizations. If propagation of a covered work results from an entity transaction, each party to that transaction who receives a copy of the work also receives whatever licenses to the work the party's predecessor in interest had or could give under the previous paragraph, plus a right to possession of the Corresponding Source of the work from the predecessor in interest, if the predecessor has it or can get it with reasonable efforts.

You may not impose any further restrictions on the exercise of the rights granted or affirmed under this License. For example, you may not impose a license fee, royalty, or other charge for exercise of rights granted under this License, and you may not initiate litigation (including a cross-claim or counterclaim in a lawsuit) alleging that any patent claim is infringed by making, using, selling, offering for sale, or importing the Program or any portion of it.

11. Patents.

A "contributor" is a copyright holder who authorizes use under this

License of the Program or a work on which the Program is based. The work thus licensed is called the contributor's "contributor version".

A contributor's "essential patent claims" are all patent claims owned or controlled by the contributor, whether already acquired or hereafter acquired, that would be infringed by some manner, permitted by this License, of making, using, or selling its contributor version, but do not include claims that would be infringed only as a consequence of further modification of the contributor version. For purposes of this definition, "control" includes the right to grant patent sublicenses in a manner consistent with the requirements of this License.

Each contributor grants you a non-exclusive, worldwide, royalty-free patent license under the contributor's essential patent claims, to make, use, sell, offer for sale, import and otherwise run, modify and propagate the contents of its contributor version.

In the following three paragraphs, a "patent license" is any express agreement or commitment, however denominated, not to enforce a patent (such as an express permission to practice a patent or covenant not to sue for patent infringement). To "grant" such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party.

If you convey a covered work, knowingly relying on a patent license, and the Corresponding Source of the work is not available for anyone to copy, free of charge and under the terms of this License, through a publicly available network server or other readily accessible means, then you must either (1) cause the Corresponding Source to be so available, or (2) arrange to deprive yourself of the benefit of the patent license for this particular work, or (3) arrange, in a manner consistent with the requirements of this License, to extend the patent license to downstream recipients. "Knowingly relying" means you have actual knowledge that, but for the patent license, your conveying the covered work in a country, or your recipient's use of the covered work in a country, would infringe one or more identifiable patents in that country that you have reason to believe are valid.

If, pursuant to or in connection with a single transaction or arrangement, you convey, or propagate by procuring conveyance of, a covered work, and grant a patent license to some of the parties receiving the covered work authorizing them to use, propagate, modify or convey a specific copy of the covered work, then the patent license you grant is automatically extended to all recipients of the covered work and works based on it.

A patent license is "discriminatory" if it does not include within the scope of its coverage, prohibits the exercise of, or is conditioned on the non-exercise of one or more of the rights that are specifically granted under this License. You may not convey a covered work if you are a party to an arrangement with a third party that is in the business of distributing software, under which you make payment to the third party based on the extent of your activity of conveying the work, and under which the third party grants, to any of the parties who would receive the covered work from you, a discriminatory patent license (a) in connection with copies of the covered work conveyed by you (or copies made from those copies), or (b) primarily for and in connection with specific products or compilations that contain the covered work, unless you entered into that arrangement, or that patent license was granted, prior to 28 March 2007.

Nothing in this License shall be construed as excluding or limiting any implied license or other defenses to infringement that may otherwise be available to you under applicable patent law.

12. No Surrender of Others' Freedom.

If conditions are imposed on you (whether by court order, agreement or otherwise) that contradict the conditions of this License, they do not excuse you from the conditions of this License. If you cannot convey a

covered work so as to satisfy simultaneously your obligations under this License and any other pertinent obligations, then as a consequence you may not convey it at all. For example, if you agree to terms that obligate you to collect a royalty for further conveying from those to whom you convey the Program, the only way you could satisfy both those terms and this License would be to refrain entirely from conveying the Program.

13. Remote Network Interaction; Use with the GNU General Public License.

Notwithstanding any other provision of this License, if you modify the Program, your modified version must prominently offer all users interacting with it remotely through a computer network (if your version supports such interaction) an opportunity to receive the Corresponding Source of your version by providing access to the Corresponding Source from a network server at no charge, through some standard or customary means of facilitating copying of software. This Corresponding Source shall include the Corresponding Source for any work covered by version 3 of the GNU General Public License that is incorporated pursuant to the following paragraph.

Notwithstanding any other provision of this License, you have permission to link or combine any covered work with a work licensed under version 3 of the GNU General Public License into a single combined work, and to convey the resulting work. The terms of this License will continue to apply to the part which is the covered work, but the work with which it is combined will remain governed by version 3 of the GNU General Public License.

14. Revised Versions of this License.

The Free Software Foundation may publish revised and/or new versions of the GNU Affero General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the Program specifies that a certain numbered version of the GNU Affero General Public License "or any later version" applies to it, you have the option of following the terms and conditions either of that numbered version or of any later version published by the Free Software Foundation. If the Program does not specify a version number of the GNU Affero General Public License, you may choose any version ever published by the Free Software Foundation.

If the Program specifies that a proxy can decide which future versions of the GNU Affero General Public License can be used, that proxy's public statement of acceptance of a version permanently authorizes you to choose that version for the Program.

Later license versions may give you additional or different permissions. However, no additional obligations are imposed on any author or copyright holder as a result of your choosing to follow a later version.

15. Disclaimer of Warranty.

THERE IS NO WARRANTY FOR THE PROGRAM, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE PROGRAM "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM IS WITH YOU. SHOULD THE PROGRAM PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION.

16. Limitation of Liability.

IN NO EVENT UNLESS REQUIRED BY APPLICABLE LAW OR AGREED TO IN WRITING WILL ANY COPYRIGHT HOLDER, OR ANY OTHER PARTY WHO MODIFIES AND/OR CONVEYS THE PROGRAM AS PERMITTED ABOVE, BE LIABLE TO YOU FOR DAMAGES, INCLUDING ANY GENERAL, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE

USE OR INABILITY TO USE THE PROGRAM (INCLUDING BUT NOT LIMITED TO LOSS OF DATA OR DATA BEING RENDERED INACCURATE OR LOSSES SUSTAINED BY YOU OR THIRD PARTIES OR A FAILURE OF THE PROGRAM TO OPERATE WITH ANY OTHER PROGRAMS), EVEN IF SUCH HOLDER OR OTHER PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

17. Interpretation of Sections 15 and 16.

If the disclaimer of warranty and limitation of liability provided above cannot be given local legal effect according to their terms, reviewing courts shall apply local law that most closely approximates an absolute waiver of all civil liability in connection with the Program, unless a warranty or assumption of liability accompanies a copy of the Program in return for a fee.

END OF TERMS AND CONDITIONS

How to Apply These Terms to Your New Programs

If you develop a new program, and you want it to be of the greatest possible use to the public, the best way to achieve this is to make it free software which everyone can redistribute and change under these terms.

To do so, attach the following notices to the program. It is safest to attach them to the start of each source file to most effectively state the exclusion of warranty; and each file should have at least the "copyright" line and a pointer to where the full notice is found.

```
    <one line to give the program's name and a brief idea of what it does.>
    Copyright (C) <year>  <name of author>

    This program is free software: you can redistribute it and/or modify
    it under the terms of the GNU Affero General Public License as published by
    the Free Software Foundation, either version 3 of the License, or
    (at your option) any later version.

    This program is distributed in the hope that it will be useful,
    but WITHOUT ANY WARRANTY; without even the implied warranty of
    MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
    GNU Affero General Public License for more details.

    You should have received a copy of the GNU Affero General Public License
    along with this program.  If not, see <https://www.gnu.org/licenses/>.
```

Also add information on how to contact you by electronic and paper mail.

If your software can interact with users remotely through a computer network, you should also make sure that it provides a way for users to get its source. For example, if your program is a web application, its interface could display a "Source" link that leads users to an archive of the code. There are many ways you could offer source, and different solutions will be better for different programs; see section 13 for the specific requirements.

You should also get your employer (if you work as a programmer) or school, if any, to sign a "copyright disclaimer" for the program, if necessary. For more information on this, and how to apply and follow the GNU AGPL, see <https://www.gnu.org/licenses/>.